

Applying AI Systems in the T&D Arena

Subrahmanyam S. Venkata¹, Zarko Sumic^{1,2},
Subramanian V. Vadari³, and Chen-Ching Liu¹

The power engineering community has capitalized on various computer technologies since the early 1960s, with most successful application to solving well-defined problems that are capable of being modeled. Although computing methods have made notable progress in the power engineering arena, there is still a class of problems that is not easy to define or formulate to apply conventional computerized methods. In addition to being difficult to express in a closed mathematical form, these problems are often characterized by the absence of one or both of the following features:

- ✧ A predetermined decision path from the initial state to goal (ill-structured problem)
- ✧ A well-defined criteria for whether an obtained solution is acceptable (open-ended problem).

Until now, these problems could not be solved with existing computerized tools. For these complex problems (such as the design of distribution facilities, load management, system protection, and system restoration) conventional computerized tools still do not exist. More often than not, these are solved using human judgment, rules of thumb, gut feeling, and other heuristic methods combined with some decision support system methods.

Over the past 10 years, rule-based expert systems have solved knowledge-intensive problems, and it is one of the few areas of artificial intelligence (AI) that have made an impact on the commercial world. Expert systems and other AI-based methodologies place emphasis on the symbolic modeling of the knowledge used during the problem-solving procedure. This is contrary to traditional computer methods used in power systems that rely on accurate models of the problem.

Power engineers have been investigating the application of AI-based methodologies to power system problems. Most of the work in the past has been geared towards the development of expert systems as an operator's aid in energy control centers for bulk power trans-

AI offers an alternative for addressing problems that are not well solved by conventional means

mission systems operating under abnormal conditions. Alarm processing, fault diagnosis, system restoration, and voltage/var control are a few key areas where significant research work has progressed to date. Results of this research have effected more than 100 prototype expert systems for power systems throughout the

United States, Japan, and Europe.

The objectives of this article are to:

- ✧ Expose engineers to the benefits of using AI methods for a host of transmission and distribution (T&D) problems that need immediate attention
- ✧ Identify problems that could be solved more effectively by applying AI approaches
- ✧ Summarize recent developments and successful AI applications in T&D.

AI-Based Methodologies

AI is a re-emerging technology that has recently attracted considerable publicity, although it has been around since the emergence of digital computers. Expert systems have gained success in medical diagnosis (MYCIN), computer hardware configuration (XCON), chemical analysis (DENDRAL), and many other industry applications, and this success has rejuvenated interest in AI research.

One simple view of AI is that it is concerned with devising computer programs to make computers smarter. In other words, the computers perform decisions like human beings with reasoning analysis. The computer programs with which AI is concerned are primarily processes involving complexity, uncertainty, and ambiguity. These are, in other words, the typical types of problem-solving and decision-making tasks that people face continually.

AI application areas are broadly classified as:

- ✧ Challenging tasks, such as machine learning, natural language processing, common-sense reasoning, and smart robots
- ✧ Formal tasks, such as game playing and theorem proving
- ✧ Expert tasks, such as engineering design, engineering and financial planning, and diagnostics.

¹ University of Washington

² Puget Sound Power & Light Company

³ ESCA Corporation

This article focuses on expert tasks in the power system area that have prospects suitable for applying AI methodologies. From this perspective, relevant AI methodologies are expert systems (ES), knowledge-based systems (KBS), and intelligent decision support systems (IDSS). Although they have striking similarities, each of these methods is used as a vehicle for different problem-solving strategies.

Other emerging computer technologies, such as artificial neural networks (ANNs), and genetic algorithms deserve consideration at the same level as the AI methodologies mentioned previously. ANNs and genetic algorithms are broadly classified under the AI domain although they do not share all the properties of ES, KBS, and IDSS. Contrary to the AI methods that try to mimic mental processes that take place during human reasoning, ANNs, in a sense, try to simulate the neural activity that takes place in the human brain. Genetic algorithms, on the other hand, are probabilistic search procedures for optimization and learning. The concepts of algorithms are derived from natural selection and natural population genetics of biology. Since both of them can be used to computerize or automate the problem-solving procedures that were not possible until now by traditional algorithmic approach, they can be mentioned under the overall umbrella of AI technologies that can be used in the T&D area.

Expert Systems

An ES is an intelligent computer program that uses knowledge and inference procedures to solve problems that are difficult enough to require human expertise for their solution. The knowledge necessary to perform at such a level and the inference procedures used can be thought of as a model of an expert in a specialized field or domain of interest.

From the programming point of view, expert systems can use the rule-based, logic-based, or heuristics-based programming paradigm. Any of these techniques can be used to solve problems that have not been automated previously. Problem solving control strategies employed by rule-based expert systems are forward chaining, backward chaining, or a combination. The forward chaining inference control strategy is intrinsic for planning and configuring problems, whereas backward chaining is well suited to diagnostic problems. The reasoning process of an expert system can be retrieved and used for explanation facilities. The replacement of manual problem solving with an automated tool is the main benefit of the ES. An inherent feature of the expert system technique is to guide a problem-solving process

The philosophy is not to replace traditional methods with AI, but to enhance them intelligently

through a prescribed computational approach.

Knowledge-Based Systems

The KBS approach goes beyond that of expert systems in that it enriches the problem-solving strategy with methods that are not originally employed by human experts. As a result, pro-

cedural optimization routines and other number crunching may be used in the solution stage. Although the search process can be different from the one used by an expert, it is still strongly dependent upon domain knowledge.

Knowledge-based systems also can be thought of as having several different experts, each of whom has specialized in a specific domain, cooperating to achieve a solution. The problem-solving control structure of the knowledge-based system is more advanced and usually employs *knowledge about knowledge* or *metaknowledge* to guide the overall solution process. Blackboard systems and agenda-based systems are the two most frequently used types of knowledge-based system architecture.

Intelligent Decision Support Systems

A decision support system (DDS) is a computerized tool derived from decision theory used to enhance the user's ability to make decisions efficiently. DSSs are intended to explore and seek alternative solutions rather than to offer the final solution. They are well suited for open-ended problems and are mainly employed in all types of planning areas. Conventional DSSs are used for retrieving, manipulating, or summarizing data in a way that assists decision makers. Some common decision support systems are spreadsheets and database management systems.

An IDSS, on the other hand, adds intelligence to an existing decision support system in order to enhance the problem-solving ability and helps maintain a broad range of knowledge about a particular domain. IDSSs are used for capturing, organizing, and reapplying knowledge including decision rules and criteria. They can evaluate the implications of alternative decisions based on users' beliefs and assumptions. In addition, the IDSS is a good way to help decision makers incorporate knowledge processing capabilities in their everyday work. Unlike the ES and the KBS, an IDSS does not attempt to discover complete solutions; instead, it plays a support role, leaving the user in charge.

AI Applications in T&D

Recent developments in AI can be explored for other power engineering areas. Based on experience in applying expert systems to several different power system problems, it is apparent that AI-based techniques could

be effectively applied to a host of problems in T&D. The philosophy is not to replace traditional methods with AI approaches, but to enhance them in an intelligent manner. Implementation of AI approaches could be cost intensive and usually cannot guarantee optimal results, only *good enough* results. Consequently, it makes sense to use AI techniques for problems not solvable by conventional methods. The practical benefits of the AI approaches, especially from the power engineering perspective, are that AI will make computers more effective even to the point of actually realizing the automation of the power system planning and operation processes.

Table 1 identifies various application areas in T&D in the context of operations, operational planning, and planning. Most of these areas have been implemented practically or investigated through prototype developments during the last decade (areas marked with asterisks (*) show potential for future investigation).

Three of the areas listed in Table 1 are featured in this article to illustrate the state of the art. The following successful applications, developed at the University of Washington, are from three different areas (operations, operational planning, and planning):

- Customer restoration and fault testing (CRAFT)
- Intelligent operational planning aid for distribution systems (IOPADS)
- Automated electrical plant design (AEPD).

(For additional details on these three applications and other areas listed in this table, readers are encouraged to peruse the For Further Reading list.)

Customer Restoration and Fault Testing

CRAFT is a rule-based system capable of locating the faulted section of a subtransmission line protected with automatic switches and power circuit breakers. Besides the fault location function, the expert system also uses logic to determine the status of unsupervised automatic and manual switches. These switches are not monitored or remote-controlled, and, therefore, the status may be

Table 1.
AI application areas for T&D problems

Operations	Operational Planning	Planning
Security assessment Unit commitment Volt/var control Alarm diagnosis Fault location and diagnosis (CRAFT) System restoration Interpretation of results from complex programs State estimation Short-term load forecasting*	Short-term load forecasting*	dc systems Relay testing ac transmission design System protection design & coordination Long-term load forecasting*
System monitoring & diagnosis Environment for operational aids Load shedding System configuration management for loss minimization System restoration Substation monitoring and control Load management control* Trouble call analysis*	Intelligent operational planning aids (IOPADS)	Underground residential distribution design (AEPD) Maintenance scheduling Safety regulation Remedial action Switching action Verification Pricing* DSG management*

* Potential areas for future AI application

unknown to dispatchers. The program suggests switching actions to restore the distribution substations served from the subtransmission line.

Each automatic switch has one or more function types, such as *open on dead line* or *close on hot line*, with a time setting for each function type. The settings of circuit breaker reclosing times and automatic switch operating times are based on logic. Consequently, the problems of fault location and switch status calculation are solved naturally with a rule-based technique. The CRAFT system is implemented in the OPS83 rule-based tool, which is very efficient in speed. The project is sponsored by Electric Power Research Institute and Puget Sound Power & Light Company (PSPL).

CRAFT has been integrated with the real-time database management system, called HABITAT, at the PSPL control center. The user interface evolved from one that is based on primitive text-based messages to the current version which, is fully consistent with the energy management system (EMS) interface. Analysis of the operational record shows 3 failures out of a total of 67 fault scenarios. All failures are due to the system's lack of information on maintenance switching, which involves

phone communication between the dispatcher and the field personnel. The PSPL EMS is being upgraded from a traditional primary-back configuration to a cluster of workstations. Continuing development will result in full compatibility in the upgraded network environment.

An algorithm for maintenance of rule-based systems has been developed and applied to the CRAFT system. This algorithm uses set relations and basic logic to detect the relations among rules, which include cause and effect, mutual exclusion, redundancy, conflict, subsumption, and implication. This proposed relation detection method utilizes an algorithm similar to the RETE inference engine used in the OPS83 tool.

Figure 1 shows typical program output results, in this case, a subtransmission line (White River Electron Heights 1), the status of the line, and the fault location suggested by CRAFT. The program suggestions are as follows:

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FAULT APPEARS TO LIE BETWEEN 981-586
OPEN 981 BY SUPERVISORY
CLOSE 412 LOCALLY AFTER AUTOMATIC
FUNCTION IS DEACTIVATED.
CLOSE 980 BY SUPERVISORY
523 WILL CLOSE WHEN THE LINE IS ENERGIZED
DO YOU WISH TO ACCEPT THE
PROPOSED SOLUTION (Y/N)?

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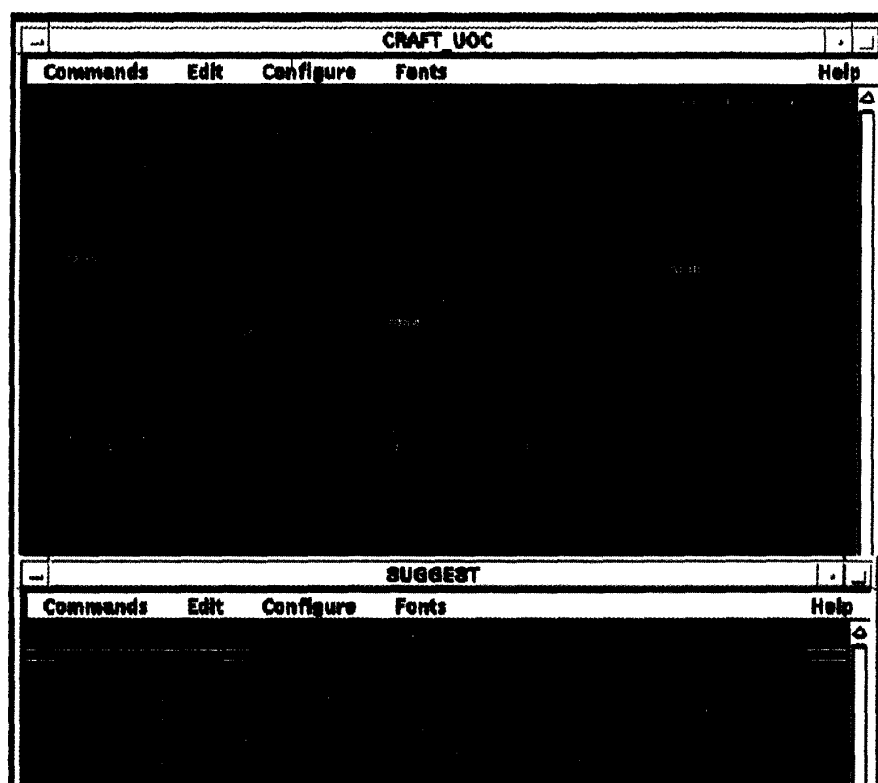


Figure 1. CRAFT output results, showing line and fault locations in a subtransmission line

Operational Planning

Distribution system planning engineers study the operating condition and feeder protection regularly. Establishing a complete and reliable database for the large number of devices in the distribution feeders is a necessary step toward computer applications. The actual use of power flow software for distribution feeders has become more common in the utilities today. Algorithms for remedial control, feeder restoration, loss minimization, and coordination of protective devices have also been developed. However, each algorithm is separated and planning engineers often find it difficult to use different programs to analyze a scenario. Another weakness in existing algorithms is that they do not provide remedial actions when violations of voltage/current limits or miscoordinations among protective devices occur.

IOPADS is an integrated software package for distribution system operational planning. The integrated system combines four algorithms: distribution power flow, loss minimization, feeder restoration, and protection coordination. Three knowledge bases have been developed. The first module is able to suggest remedial control actions to eliminate voltage or current violations. The second module provides the necessary changes in protective devices in order to

eliminate miscoordination in the feeders. The last module is related to the use of feeder reconfiguration in loss minimization or feeder restoration. Constraints are generated in order to avoid violation of operating limits in the reconfiguration. These constraints help to reduce the amount of search. IOPADS has been implemented in a real-time expert system environment, G2. In Figure 2, the G2 shell provides windows that report an outage that requires a restorative switching action suggested by IOPADS. IOPADS also decides that this action leads to miscoordination problems between protective devices. The necessary remedial adjustment for the protective device is also shown.

Electrical Plat Design

The design of the underground residential distribution (URD) system for new developments (plat design) is

an everyday task for electric utility engineers. The design is based on heuristic strategies and rules of thumb based on the experience gained from previous designs. The lack of an optimizing algorithm for the overall design indicates that the AI approach might be worthwhile to explore. A geographic information system (GIS) is a software environment for managing geographic features, and this can be used for automated data preparation and generation of the attributes that are required for automated URD design.

PSPL and University of Washington have jointly developed the AEPD system as a tool for the computerized URD design. To mimic the spatial reasoning needed to perform URD design routing/siting tasks, a KB-GIS environment was developed that integrates a GIS (ARC-INFO) with an expert system shell (NEXPERT-OBJECT). To automate the URD design, the system uses IDSSs as an appropriate AI methodology. To accommodate diverse software environments, data models, and problem-solving paradigms, the AEPD tool utilizes the blackboard architecture shown in Figure 3. This architecture consists of different problem-solving components called knowledge sources as well as the human designer. In such a hybrid environ-

ment, the principal task of the GIS is to act as a data repository and structure and formalize a *real world* representation required for spatial and engineering reasoning emulated by various knowledge sources.

In Figure 4, comparison of the manual and automated designs shows a close agreement. The main difference is that a designer needs 2 to 3 days to complete this task manually, while one design solution can be reached in 3 minutes with the help of AEPD. Since the time required to provide a solution is drastically reduced, AEPD can be used effectively as an IDSS tool to provide distribution designers with an automated way to explore the impact of different design scenarios.

Benefits and Limitations

This article highlights AI techniques and applications that are being implemented to help solve T&D problems. It should be made clear that AI methodologies are not a panacea for solving all power problems, but they have legitimate applications in automating repetitive, intellectual tasks. They can provide a degree of

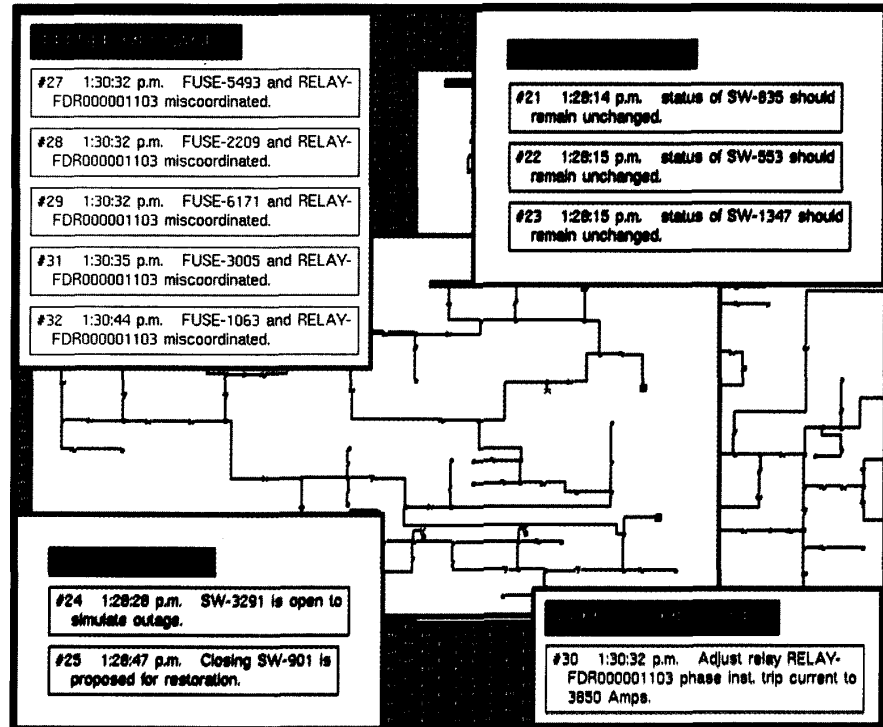


Figure 2. IOPADS output results, showing an outage and switching action in a distribution feeder

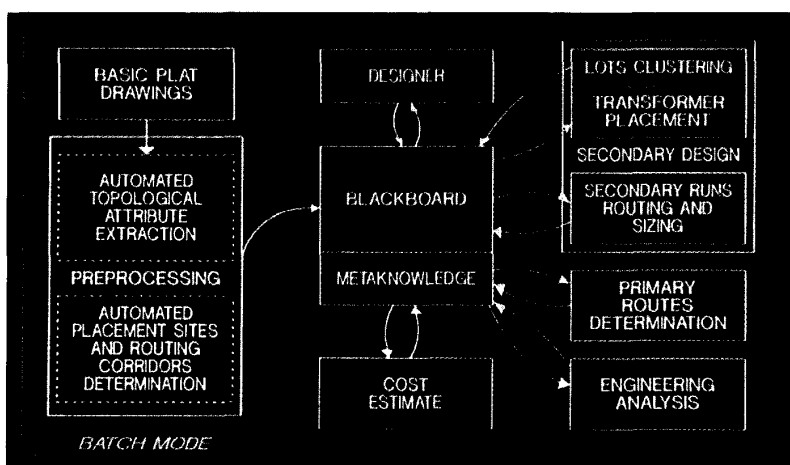


Figure 3. Blackboard-based AEPD problem-solving architecture

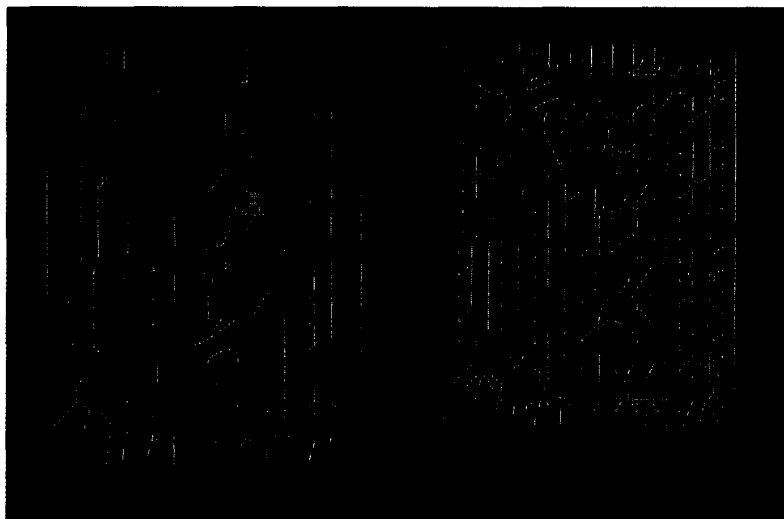


Figure 4. Manual (left) vs automated (right) plot design results

consistency often missing in human decision making endeavors.

In recent years, many industries, including electric utilities, have encouraged early retirements to become more competitive, which may result in shortage of personnel and loss of expertise. One remedy is to capture the knowledge of retiring and transferring engineers and store it in AI programs to help train younger engineers.

Like any technology, AI is not without its limitations. These AI methods could be costly; they have a basic inability to *learn* in the commonly accepted sense of learning; therefore, they require continual maintenance to stay up-to-date. However, in the final analysis, power engineers are encouraged to keep AI in mind as one option for solving many problems which are not well solved by conventional means.

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For Further Reading

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Biographies

S.S. Venkata received his BSEE and MSEE degrees from India and his PhD from the University of South Carolina, Columbia. He taught at the University of Lowell and West Virginia University before joining the University of Washington, Seattle, in 1979. He is a professor of electrical engineering and director of the Electric Energy Industrial Consortium. Dr. Venkata is a Fellow of the IEEE, a member of Tau Beta Pi, Sigma Xi, and Eta Kappa Nu. He has published and presented more than 90 papers and is a coauthor of

Introduction to Electric Energy Devices (Prentice-Hall, 1987).

Zarko Sumic received his BSEE (Dipl Eng) from the University of Split, Croatia, MSEE from the University of Zagreb, Croatia, and PhD from the University of Washington, Seattle. He was a faculty member at the University of Split from 1976 to 1988. He is a staff engineer with the Puget Sound Power & Light Company, Bellevue, Washington, working on GIS and AI application in distribution planning and operation. Dr. Sumic is also an affiliate assistant professor of electrical engineering at the University of Washington, Seattle. He is a senior member of the IEEE and a member of CIGRE and Eta Kappa Nu. He has published and presented more than 30 papers.

Subramanian V. Vadari received his BSEE from Osmania University, India and his MSEE and PhD from the University of Washington at Seattle. Since 1987, he has been with ESCA Corporation, Bellevue, Washington, where he is currently leading the development of Dispatcher Training Simulator Applications. Dr. Vadari is a member of IEEE, Tau Beta Pi, and Eta Kappa Nu.

Chen-Ching Liu received his BSEE and MSEE from National Taiwan University and his PhD from the University of California, Berkeley. He was an instructor at Army Signal and Electronics School, Taiwan, from 1978 to 1980. Since 1983, he has been with the University of Washington, where he is a professor of electrical engineering. He is a senior member of the IEEE and chairs the PES Power System Engineering Committee Working Group on Artificial Intelligence Applications. He also chairs a CIGRE Task Force, TF 38-06-03, on expert system applications to power systems.